

All measurements are in micro meter

Image 1_3 T Spen 820.png
Root Length (um) Width (um)
Main Stem 1472.1
Density 0

Image 1_4 T pi39 420.png
Root Length (um) Width (um)
Main Stem 1386.1
1 315.8 19.5
2 115.7 18.5
3 115 18.4
4 74.2 16.2
5 267 19
6 234.6 19
7 136.3 18.8
8 159.6 17.4
9 236.8 18.8
10 224.3 16.1
11 175.8 19
12 308.2 18.7
13 123.3 18
Density 0.009379

Image 2_3 L LODA 620.png
Root Length (um) Width (um)
Main Stem 1517.3
1 75.7 11.7
2 197.2 12
3 61.9 10.8
4 12.3 14
5 10.6 7.3
6 40.2 11.3
7 62.3 11.3
8 101.1 11.4
9 11.7 9.1
10 100.7 12.8
11 44.5 11.7
12 34.2 13.4
13 21.7 12.7
14 12.2 3.3
15 64.1 12.5
16 82.3 11
17 45.7 11.6

18	151.2	11.8
19	62.5	12.3
20	153.8	12.9
21	81.6	12.2
22	84.4	13
23	57.3	13.5
24	100.4	11.4

Density 0.015818

Image 2_3 L Spen 820.png
 Root Length (um Width (um)

Main Stem	1321.4	
1	210.2	10.6
2	262.3	13.3
3	87.9	11.6
4	281.9	14.6
5	263.4	10.9
6	148.7	12
7	104.2	13.9
8	20.9	12.3
9	234	13.6
10	262.2	11.7
11	213.3	10.1
12	68.2	10.4
13	271.6	13
14	148.5	10.9
15	272.3	12.3
16	147.9	12.2
17	207.5	12.5
18	214.8	10.3
19	161.2	9.8
20	267.7	12.6

Density 0.015136

Image 2_5 T Loda 420.png
 Root Length (um Width (um)

Main Stem	1253	
1	32.6	9.7
2	51.2	12.8
3	94.1	14.7
4	84.5	11.8
5	62	9.4
6	121.9	13.6
7	106.5	11.2

8	117.8	12.4
9	9.6	8.9
10	67.2	12.5
11	86.5	7.9
12	50.4	10.4
13	19.1	11.8
14	174	14.3
15	96.5	12.8
16	44.9	10.8
17	94.7	14.2
18	138.4	12
19	65.3	11.9
20	65.9	9.8
21	31.7	11.7
22	79	13.2
23	152.1	10.2
24	25.5	11.7
25	110.2	11.8
26	166.5	9.9
27	19.8	9.7

Density 0.021548

Image	2_5 T Spen 420.png	
Root	Length (um)	Width (um)
Main Stem	1598.2	
1	81.5	11.3
2	99.5	11.6
3	55.9	9.4
4	19.2	8.4
5	40.2	9.8
6	29.3	11.1
7	63.3	9.5
8	45.8	9
9	38.5	7.7
10	98	10.6
11	98.1	12.1
12	43.1	9.1
13	41.3	11
14	23	8.4
15	103.4	11.4
16	48.4	7.2
17	71.4	11.1
18	92.4	12.7
19	20.3	9.8

20	54.7	9.8
21	43.6	11.9
22	40.2	7.4
23	45	9.5
24	100.8	7.3
25	67.8	12.4
26	45.4	9.2
27	76.6	10.3
28	48	9.3
29	95.5	9.2
30	67.5	12
31	72.1	9.7
32	128	10.8
33	54.7	14.5
34	46.3	7.4
35	65.9	7.2
36	65.6	9.3
37	42.6	9.3
38	68.5	11.2
39	63.2	11.3
40	44.8	3.5
41	14.2	10.2
42	41.8	2.5
43	42.3	11.4

Density 0.026905

Image	3_3 L Spen 420.png	
Root	Length (um)	Width (um)
Main Stem	1518.6	
1	67	11.1
2	17.6	11.8
3	80.1	10.8
4	79.5	12.6
5	107.9	13
6	122.4	11.8
7	19.4	13.4
8	66.9	12.8
9	133.2	12.9
10	12.8	3.7
11	34.2	11.2
12	59.3	15.8

Density 0.007902

Image 3_3 L lod 420.png

Root	Length (um	Width (um)
Main Stem	1325.6	
1	188.6	12.9
2	168.4	15.4
3	103.7	17.1
4	232.4	15.3
5	198.4	12.5
6	110.9	10.1
7	313.2	14.7
8	144.9	12.8
9	274.3	16.2
10	131.4	14.1
11	175.5	12.6
12	78	13.2
13	259.2	15.1
14	254.1	16.7
15	172.6	13.5
16	119.7	13.7
17	108.4	12.3
18	134.7	14.7
19	222	14.5
20	271.6	15.7
21	171.3	16.3
Density	0.015842	

Image	3_5 T LODA 620.png	
Root	Length (um	Width (um)
Main Stem	1453.9	
1	166.4	14.2
2	120.9	10.3
3	71.5	11.6
4	30.3	9.1
5	67.5	12.9
6	127	12
7	63.7	11.1
8	65.9	11
9	88.8	11.2
10	112.7	11.5
11	70	10
12	55.8	10.9
13	63.1	13.6
14	88.9	11.7
15	36.8	11.4
16	47.4	13.6

17	32.8	14.3
18	111.7	11
19	28.9	11.9
20	92.9	11.7

Density 0.013756

Image 3_5 T pi39 420.png

Root Length (um Width (um)

Main Stem 1199.2

1	123.6	17.4
2	233.2	19
3	70.3	14.8
4	142.9	19.1
5	68.3	13
6	533.1	18.2
7	75.7	14.8

Density 0.005837

Image 4_1 L CLARK 820.png

Root Length (um Width (um)

Main Stem 1378.6

1	295.8	11.5
2	177.6	13.5
3	261.5	11.6
4	263.9	13.1
5	252.1	10.7
6	147	10.9

Density 0.004352

Image 4_1 T CLARK 820.png

Root Length (um Width (um)

Main Stem 1609.6

Density 0

Image 4_1 T Spen 420.png

Root Length (um Width (um)

Main Stem 1204

1	50.8	10.5
2	197.4	9.8
3	200.2	11.9
4	149.6	10.7
5	110.2	15.9
6	319.3	12.6
7	19.2	5.6

8	99.3	12.1
9	62.3	10.6
10	59.3	10.2
11	168	11.1
12	41	9.7

Density 0.009967

Image 4_1L Clark 420.png
 Root Length (um) Width (um)

Main Stem	1510	
1	249.1	14.7
2	199.9	9.2
3	42.9	9.4
4	287.5	12.4
5	401.2	13.3
6	82.2	11.4
7	27.1	10.4
8	386.8	11.4
9	521.6	13.3
10	69	11

Density 0.006623

Image 4_2 L loda 820.png
 Root Length (um) Width (um)

Main Stem	1390.3	
1	158.8	10.3
2	248.6	11.4
3	151.5	11.3
4	85.3	13.7
5	76.5	11.1
6	66.8	7.4
7	347	12.9
8	346.7	13.8
9	218.9	13.1
10	173.5	12.1
11	188.7	10
12	167.9	11
13	230.3	15
14	192.1	11.8
15	422.5	15.1
16	28.2	10.6
17	256.7	11.6
18	332.8	14
19	363.8	12.5

20	246.9	13.3
Density	0.014386	

Image	4_2 T PI39 820.png
Root	Length (um) Width (um)

Main Stem	1309.9	
1	209.1	12.7
2	287	12.7
3	38.8	17.9
4	79.9	11.4
5	170.2	12.1
6	127.9	14.1
7	121.4	13.4
8	160.1	15.6
9	82.8	11.3
10	49.1	8.6
11	171.6	13
12	206.1	11.5
13	165.5	15.3
14	168.4	15.2
15	152.8	13.6
16	161.8	12

Density	0.012214
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Image	4_2 T pi39 620.png
Root	Length (um) Width (um)

Main Stem	1350.8	
1	31.2	11.8
2	48.2	12.3
3	17.8	7.9
4	42.4	11.5
5	36.5	13.4
6	43.2	11.9
7	0	0
8	9.1	7.6
9	11.1	7.1
10	59.4	11.6
11	60	10.5

Density	0.008144
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Image	4_3 L Spen 420.png
Root	Length (um) Width (um)

Main Stem	1482.9	
1	58.8	12.4

2	46	6.1
3	32.4	9.5
4	35.9	12.1
5	81.1	13.5
6	163.3	15.4
7	58.1	5.5
8	80.5	13.8
9	68.7	12.9
10	30.1	11.5
11	183.8	16.2
12	90.3	17.4
13	98.1	13.7
14	69.9	15.1
15	275	14.4
16	55.2	12.2
17	22	16.2
18	122.5	13.1
19	19.4	9.7
20	64.8	13
21	32.4	10.9

Density 0.014161

Image 4_3 L loda 420.png

Root Length (um) Width (um)

Main Stem 1323.2

1	129.7	11.5
2	144.6	13.2
3	119.7	14.2
4	102.4	16.3
5	102	12.8
6	151.7	12.8
7	159.1	15.6
8	108.2	15
9	252.8	13.3
10	132.3	14.1
11	176.1	11.6
12	250	15.3
13	199.7	9.5
14	117.7	14.7
15	129.1	12.9
16	222.1	12.4
17	308	13.3
18	157.4	12.6
19	155.4	15

20	233.8	13.5
21	215.9	13.6
22	225	17.1
23	196.4	10.3
24	172.9	14.9

Density 0.018137

Image 4_3 T HS93 820.png
 Root Length (um Width (um)

Main Stem	1203.8	
1	26.8	11.1
2	28.8	12.2
3	27.8	8.6
4	21.9	9.5
5	19.8	5.2
6	31	9.1
7	0	0

Density 0.005815

Image 4_4 L CLARK 820.png
 Root Length (um Width (um)

Main Stem	1230.9	
1	129	13.5
2	156	12.4
3	85.9	12.3
4	139.5	13
5	29.3	13.5
6	97.5	9.6
7	71.8	16.2
8	217.6	13.4
9	142.5	11.8
10	130.6	11.8
11	144.6	12.8
12	129.4	10.8
13	529.4	12.2
14	149.9	14.3
15	167.3	11.7

Density 0.012186

Image 4_4 L PI39 820.png
 Root Length (um Width (um)

Main Stem	1489.1	
1	302.2	10
2	445.1	13.3

3	28.6	4.7
4	39.9	11.3
5	518.9	13
6	425.6	12.5
7	354.7	12.5
8	256.8	9.2
9	39.3	11.3
10	278.9	11.7
11	39.4	12.4
12	74	12.6

Density 0.008059

Image 4_4 T LODA 620.png

Root Length (um Width (um)

Main Stem	1251.4	
1	62.3	12.7
2	62.4	9
3	33	3.6
4	170.7	11.6
5	17.8	5.7
6	116.3	12.6
7	0	0
8	38.4	10.9
9	111.1	12.5
10	132.4	13.6
11	87.8	12.9
12	119.2	11.3
13	77.1	8.5
14	78.7	12.9
15	88.9	10.8
16	30.4	10.9
17	122.4	12.4
18	99.4	15.6
19	125.3	14.3
20	102.8	9.3
21	29.4	8
22	51.8	7.3
23	81	12.7
24	30.4	10.9
25	77.8	11.7
26	98.9	12.8
27	101.8	10.3
28	87.4	12
29	98.6	10.5

30	77.2	10
Density	0.023973	

Image	4_5 L HS93 820.png
Root	Length (um) Width (um)

Main Stem	1438.3	
1	244.7	13.4
2	51.4	13.3
3	55.1	15.7
4	315.3	11.7
5	146.4	10.8
6	74.1	14.4
7	3.9	7.8
8	59	11.8
9	187.3	16.2
10	47.5	16.3
11	181	9.6
12	62.9	14.3
13	3.4	0.7
14	96.9	15.9
15	32.1	11.1
16	189	12.2
17	24.4	14.4
18	0	0
19	42.6	10
20	222.9	11.4
21	239.7	13
22	10.8	8.3
23	368.2	16.2
24	60.1	14.2
25	184.4	12.6
26	53.6	11.8
27	55.5	15.5
28	72.2	13
29	142.3	11
30	110.2	8.1
31	526	13.7
32	110.6	7.3
33	103.2	13.1
34	161.5	9.6

Density	0.023639
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Image	4_5 L hs93 420.png
Root	Length (um) Width (um)

Main Stem	1280.7	
1	588.4	15.4
2	585.9	15.6
3	291.5	11.1
4	585.7	15.2
5	376.4	12.7
6	550.1	16.6
7	324.6	18.8
Density	0.005466	

Image	4_5 L pi39 420.png	
Root	Length (um)	Width (um)
Main Stem	1209.2	
1	181.1	11
2	63.9	12.5
3	162.2	10.9
4	250.8	8.7
5	389.7	13.7
6	252.5	12
7	231.9	10.6
8	405.7	10.3
9	279.2	11.2
10	397.6	10.6
11	0	0
12	443	11.3
13	283	12.5
14	47.9	9.9
Density	0.011577	

Image	4_5 T CLARK 820.png	
Root	Length (um)	Width (um)
Main Stem	1524.4	
Density	0	

Image	4_5 T Spen 420.png	
Root	Length (um)	Width (um)
Main Stem	1334	
1	128.3	17.7
2	227.1	17.6
3	76	16.9
4	136	14.2
5	258.8	17.5
6	217.7	17.5
7	194.2	18.7

8	50.9	17.2
9	47.3	14.7
10	179.4	18.2
11	144.3	16.9
12	297.6	18.5
13	38.1	14
14	207.8	18.3
15	168.4	16.6
16	160.7	16.3

Density 0.011994

Image 4_5 T pi39 420.png
Root Length (um) Width (um)
Main Stem 1203.4

1	63.8	19
2	583.1	19.4
3	496.4	18.1
4	650.4	19
5	115.6	19.5
6	96.5	17.6

Density 0.004986